

POKHARA UNIVERSITY

Level: Bachelor
Programme: BCSIT
Course: Programming Language

Semester: Fall

Year: 2024
Full Marks: 100
Pass Marks: 45
Time: 3 hrs.

Candidates are required to answer in their own words as far as practicable. The figures in the margin indicate full marks.

Section "A"

Very Short Answer Questions

Attempt all the questions. [10×2=20]

1. Differentiate Between Compiler and Interpreter.
2. What is a keyword in C? Give two examples.
3. Explain the difference between getchar() and scanf() functions.
4. What is the purpose of the sizeof operator in C?
5. Explain formatted input/output in c.
6. How do you declare a 2D array in C? Provide an example.
7. What is recursion?
8. Explain the difference between malloc() and calloc().
9. What are the benefit of using structure in programming?
10. Difference between sequential and random access in file handling.

Section "B"

Descriptive Answer Questions

Attempt **any five** questions. [5×10=50]

11. Explain different types of problem-solving techniques used in programming. What are the differences between flowcharts and algorithms?
12. Explain the different types of operators in C with suitable examples.
13. Write a program to check if a given number is prime or not.
14. Write a C program to find the factorial of a number, recursively.
15. Explain between passing a variable by value and passing a variable by reference.
16. Write a program to input numbers into an array and sort them in ascending order.

Group "C"

Long Answer Questions

Attempt **any two** questions. [2×15=30]

17. a. Differentiate between a union and structure with example. [5]
b. Write a Program: [10]
 - i. Create an array of student for 5 students
 - ii. To calculate and display the average the mark of student

Each student has the following attributes:

- Roll number (integer)
- Name (string)
- Marks in 5 Subjects.

18. a. Write a program to do matrix multiplication. Check and verify the size of matrix before allocating memory dynamically. [8]
b. Write a program to find given word is palindrome or not? [7]
19. a. Write a program to manage employee records using file handling: Read details (name, department, salary) of a total of 5 employees from the console/terminal and save the details in a file named 'employees.txt'.
b. Retrieve and display records of all employees whose salary is greater than Rs. 50,000.